

Dirac algebra with feynsage

*Traces, spin sums, photons and colour,
written the way the QFT book writes them*

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Source code at github.com/aburousan/feynsage

WHAT THIS GUIDE IS

In a first QFT course we spend many hours on traces of gamma matrices. We write $\text{tr}(\not{p}\gamma^\mu \not{q}\gamma^\nu)$, use the trace theorems, contract indices, sum over spins and put in the kinematics. This guide shows how to let the computer do all of that, while we still write everything as in the book.

The package feynsage runs inside SageMath. In it you type `slash(p) * gamma(mu) * slash(q) * gamma(nu)` and ask for the trace. A program called FORM does the algebra in the background and the answer comes back as a normal Sage formula. You never write a line of FORM.

The guide starts from zero. First the new symbols, then traces, contractions, γ^5 and spinors. Then seven full applications. These are $e^+e^- \rightarrow \mu^+\mu^-$ up to the total cross section, Compton scattering up to the Klein-Nishina formula, left-handed currents, colour factors of QCD with the ratio R , a massive vector boson, the equivalent photon approximation and the $g-2$ of the electron from a new scalar or pseudoscalar (problems 5.5, 6.2 and 6.3 of Peskin and Schroeder). Every result is checked against the textbook.

HOW TO READ THE BOXES

Every grey box marked **In [n]** is real code. All boxes were run in this order in one SageMath session, like a Jupyter notebook. The box under it marked **Out [n]** is what came out. The formulas in the output boxes are typeset from the LaTeX that feynsage itself writes. Nothing in an output box was typed by hand.

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1 Before we start

1.1 What you need

- **SageMath** 10.7 or newer. It is free.
- **FORM**, the program that does the algebra. It is also free.
- The package feynsage from github.com/aburousan/feynsage.

On macOS, Linux or Windows (WSL) one script does everything:

```
git clone https://github.com/aburousan/feynsage.git
cd feynsage
./install.sh          # finds or installs SageMath and FORM, then installs feynsage
```

After that open a notebook with `sage -n jupyterlab` and choose the SageMath kernel. Every cell of this guide can be run again with `sage docs/qft_guide/run_cells.sage`.

1.2 What FORM is and why we do not write it

FORM was written by Jos Vermaseren for very long algebra, the kind that appears in particle physics. It is fast. A trace of twelve slashed momenta has 10395 terms in d dimensions and FORM finishes it in about 14 ms. But FORM has its own language, with lines like `g_(1,p,mu,q,nu);` and `trace4,1;`. Learning it takes time that a student would rather spend on physics.

So feynsage stands in the middle. You write Python and Sage in the notation of the book. feynsage finds out which symbols are momenta and which are indices, writes the FORM program, runs it and reads the answer back into Sage. If you are curious, Section 17 shows how to see the FORM program it wrote.

1.3 Conventions in one place

We use the conventions of Peskin and Schroeder [1]. The metric is $g^{\mu\nu} = \text{diag}(1, -1, -1, -1)$. The gamma matrices obey $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$ and

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$$

which is their eq. (3.68) on p. 50. With this choice their eq. (5.5) on p. 134 says

$$\text{tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^5) = -4i\epsilon^{\mu\nu\rho\sigma}$$

and together these fix $\epsilon^{0123} = +1$. A trace of the unit matrix is 4. Upper and lower indices are not told apart in the code. A repeated index means a sum, with $g_{\mu\nu}$ put in as in the book.

2 The new symbols

All new names come with one line, `from feynsage import *`. The table is the whole vocabulary of this guide. The first column is what you type, the second is what the book writes.

you type	in the book	what it is
<code>p, q = momenta("p q")</code>	p, q	four-momenta
<code>mu, nu = lorentz_indices("mu nu")</code>	μ, ν	Lorentz indices
<code>gamma(mu)</code>	γ^μ	Dirac matrix
<code>slash(p)</code>	$\not{p} = \gamma^\mu p_\mu$	slashed momentum
<code>slash(p) + m</code>	$\not{p} + m$	propagator numerator
<code>gamma5()</code>	γ^5	
<code>PL(), PR()</code>	$(1 - \gamma^5)/2, (1 + \gamma^5)/2$	chiral projectors
<code>sigma(mu, nu)</code>	$\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$	

you type	in the book	what it is
<code>one()</code>	$\mathbb{1}$	unit matrix
<code>u(p, m), v(p, m)</code>	$u(p), v(p)$	spinors (columns)
<code>ubar(p, m), vbar(p, m)</code>	$\bar{u}(p), \bar{v}(p)$	barred spinors (rows)
<code>dot(p, q)</code>	$p \cdot q$	scalar product
<code>comp(p, mu)</code>	p^μ	component
<code>metric(mu, nu)</code>	$g^{\mu\nu}$	metric
<code>epsilon(mu, nu, rho, tau)</code>	$\varepsilon^{\mu\nu\rho\tau}$	Levi-Civita tensor
<code>polarization("e", k)</code>	$\varepsilon(k)$	photon polarization
<code>color_indices("a b")</code>	a, b	gluon colour indices
<code>quark_colors("i j")</code>	i, j	quark colour indices
<code>T_color(a, i, j), f_color(a, b, c)</code>	$(T^a)_{ij}, f^{abc}$	SU(N) generators and structure constants

And the things you do with them:

function	what it does
<code>dirac_trace(expr)</code>	the trace $\text{tr}(\dots)$, done by FORM
<code>contract(expr)</code>	sums every repeated index, also inside a line of gamma matrices
<code>conjugate(M)</code>	the complex conjugate M^* of an amplitude
<code>spin_sum(M * conjugate(M))</code>	$\sum_{\text{spins}} M ^2$
<code>polarization_sum(expr, e)</code>	$\sum_{\text{pol}} \varepsilon^\mu \varepsilon^{*\nu} \rightarrow -g^{\mu\nu}$
<code>color_factor(expr, N=3)</code>	sums every repeated colour index
<code>.subs({dot(p, p): m^2})</code>	puts in kinematics (ordinary Sage)

When a result is printed as text you see the names `Dot(p, q)`, `Comp(p, mu)`, `Metric(mu, nu)` and `Epsilon(...)`. In a notebook `show(...)` prints them as $p \cdot q$, p^μ , $g^{\mu\nu}$ and $\varepsilon^{\dots}$, the way this guide shows them.

3 First steps

Load feynsage:

```
In [1]: from feynsage import *
```

Now tell Sage which symbols are momenta and which are indices. This is the only bookkeeping you do.

```
In [2]: p, q, k = momenta("p q k")
        mu, nu, rho, tau = lorentz_indices("mu nu rho tau")
        m = var('m')
```

`m` is an ordinary Sage symbol, here a mass. Now we can write a product of Dirac matrices exactly as in the book. A mass is added to a slash in the normal way.

```
In [3]: slash(p) * gamma(mu) * (slash(q) + m)
```

```
Out[3]: | 
$$\not{p} \gamma^\mu (\not{q} + m)$$

```

Gamma matrices do not commute, so feynsage keeps them in the order you write. $\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu$ is not zero:

```
In [4]: gamma(mu) * gamma(nu) - gamma(nu) * gamma(mu)
```

```
Out[4]: | 
$$\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu$$

```

GAMMA ALSO MEANS THE GAMMA FUNCTION

Sage already has a function called `gamma`, Euler's $\Gamma(x)$. feynsage keeps it. If the argument is a Lorentz index you get γ^μ , otherwise you get Γ .

```
In [5]: gamma(5), gamma(1/2)
```

```
Out[5]: (24, sqrt(pi))
```

If you misspell an index, say `gamma(muu)`, feynsage notices it as soon as you take a trace and tells you to declare `muu` (Section 18).

4 Dirac traces

The simplest trace, $\text{tr}(\gamma^\mu \gamma^\nu) = 4g^{\mu\nu}$:

```
In [6]: dirac_trace(gamma(mu) * gamma(nu))
```

```
Out[6]: 4 g^{\mu\nu}
```

Four gamma matrices give the famous three terms of Peskin and Schroeder eq. (5.5):

```
In [7]: dirac_trace(gamma(mu) * gamma(nu) * gamma(rho) * gamma(tau))
```

```
Out[7]: 4 g^{\mu\tau} g^{\nu\rho} - 4 g^{\mu\rho} g^{\nu\tau} + 4 g^{\mu\nu} g^{\rho\tau}
```

A trace of an odd number of gamma matrices vanishes:

```
In [8]: dirac_trace(gamma(mu) * gamma(nu) * gamma(rho))
```

```
Out[8]: 0
```

Slashed momenta work the same way. $\text{tr}(\not{p}\not{q}) = 4p \cdot q$:

```
In [9]: dirac_trace(slash(p) * slash(q))
```

```
Out[9]: 4 (p \cdot q)
```

Mixed indices and momenta. The answer has components like p^μ :

```
In [10]: dirac_trace(slash(p) * gamma(mu) * slash(q) * gamma(nu))
```

```
Out[10]: 4 p^\nu q^\mu + 4 p^\mu q^\nu - 4 (p \cdot q) g^{\mu\nu}
```

With masses. The terms with one m have an odd number of gamma matrices and drop out by themselves:

```
In [11]: dirac_trace((slash(p) + m) * gamma(mu) * (slash(q) + m) * gamma(nu))
```

```
Out[11]: 4 m^2 g^{\mu\nu} + 4 p^\nu q^\mu + 4 p^\mu q^\nu - 4 (p \cdot q) g^{\mu\nu}
```

FEYNSAGE FUNCTION `dirac_trace(expr, dim=4, rules=None)`

Takes a product (or a sum of products) of Dirac matrices and gives its trace as a Sage expression. Every repeated index is summed. `dim` is the number of dimensions (Section 6.2), `rules` puts in kinematics before the answer comes back (Section 5). It saves you the trace theorems and the index bookkeeping.

The same thing can be written as a method:

```
In [12]: expr = slash(p) * slash(q) * slash(p) * slash(q)
          expr.trace()
```

```
Out[12]: 8 (p \cdot q)^2 - 4 p^2 q^2
```

Many traces can be done in one call. This is the fastest way when you have a long list, because FORM is asked only once:

```
In [13]: dirac_trace([gamma(mu) * gamma(mu), slash(p) * slash(p), 7])
Out[13]: [16, 4 p^2, 28]
```

5 Scalar products and kinematics

`dot` is linear in each momentum and symmetric:

```
In [14]: dot(p + k, q)
Out[14]: (k · q) + (p · q)
```

```
In [15]: dot(q, p) - dot(p, q)
Out[15]: 0
```

Scalar products are normal Sage objects, so kinematics goes in with `.subs`. Take $\text{tr}(\cancel{p}\cancel{q}\cancel{p}\cancel{q})$ with $p^2 = m^2$, $q^2 = 0$ and $p \cdot q = s/2$:

```
In [16]: s = var('s')
r = dirac_trace(slash(p) * slash(q) * slash(p) * slash(q))
r.subs({dot(p, p): m^2, dot(q, q): 0, dot(p, q): s/2})
Out[16]: 2 s^2
```

For long expressions it is faster to give the same dictionary to FORM, with `rules=`. Then FORM puts in the kinematics and only the short answer comes back:

```
In [17]: dirac_trace(slash(p) * slash(q) * slash(p) * slash(q),
                    rules={dot(p, p): m^2, dot(q, q): 0, dot(p, q): s/2})
Out[17]: 2 s^2
```

A rule can also replace a momentum. This is how momentum conservation $k_2 = p_1 + p_2 - k_1$ goes in. The rule works everywhere, also inside a slash and in a denominator:

```
In [18]: p1, p2, k1, k2 = momenta("p1 p2 k1 k2")
dirac_trace(slash(k2) * slash(p1), rules={k2: p1 + p2 - k1})
Out[18]: -4 (k1 · p1) + 4 p1^2 + 4 (p1 · p2)
```

6 Contracting indices

`contract` sums repeated indices. $g^{\mu\nu} p_\mu q_\nu = p \cdot q$:

```
In [19]: contract(metric(mu, nu) * comp(p, mu) * comp(q, nu))
Out[19]: (p · q)
```

A trace contracted with $k_\mu k_\nu$:

```
In [20]: T = dirac_trace(slash(p) * gamma(mu) * slash(q) * gamma(nu))
contract(T * comp(k, mu) * comp(k, nu))
Out[20]: 8 (k · p)(k · q) - 4 k^2 (p · q)
```

The Levi-Civita tensor. Peskin and Schroeder eq. (5.6) on p. 134 says $\varepsilon^{\alpha\beta\gamma\delta}\varepsilon_{\alpha\beta\gamma\delta} = -24$ and $\varepsilon^{\alpha\beta\gamma\mu}\varepsilon_{\alpha\beta\gamma\nu} = -6\delta^\mu_\nu$. Here are both, the second one with a momentum k in the last slot:

```
In [21]: contract(epsilon(mu, nu, rho, tau) * epsilon(mu, nu, rho, tau))
```

```
Out[21]: | -24
```

```
In [22]: contract(epsilon(tau, nu, rho, mu) * epsilon(tau, nu, rho, k))
```

```
Out[22]: | -6 k^\mu
```

6.1 Lines of gamma matrices without a trace

`contract` also works on an open product of gamma matrices. It uses $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$ to bring the line to a fixed order and then sums the repeated indices. These are the contraction identities (5.9) on p. 135 of Peskin and Schroeder:

```
In [23]: contract(gamma(mu) * gamma(nu) * gamma(mu))
```

```
Out[23]: | -2\gamma^\nu
```

```
In [24]: contract(gamma(mu) * gamma(nu) * gamma(rho) * gamma(mu))
```

```
Out[24]: | 4 g^{\nu\rho}
```

```
In [25]: contract(gamma(mu) * slash(p) * slash(q) * gamma(mu))
```

```
Out[25]: | 4 (p \cdot q)
```

The third one is $\gamma^\mu \not{p} \not{q} \gamma_\mu = 4p \cdot q$, the same as the second with momenta in place of indices.

6.2 Other dimensions

In dimensional regularization we work in d dimensions, where $\gamma^\mu \gamma_\mu = d$. Give `dim=d`:

```
In [26]: d = var('d')
          contract(gamma(mu) * gamma(nu) * gamma(mu), dim=d)
```

```
Out[26]: | (-d + 2)\gamma^\nu
```

```
In [27]: dirac_trace(gamma(mu) * gamma(nu) * gamma(mu) * gamma(nu), dim=d)
```

```
Out[27]: | -4 d^2 + 8 d
```

At $d = 4$ the first is $-2\gamma^\nu$ again and the second is -32 , the same as $\text{tr}(-2\gamma^\nu \gamma_\nu) = -2 \cdot 16$.

7 Gamma5 and chirality

The trace with γ^5 of Peskin and Schroeder eq. (5.5):

```
In [28]: dirac_trace(gamma(mu) * gamma(nu) * gamma(rho) * gamma(tau) * gamma5())
```

```
Out[28]: | -4i \epsilon^{\mu\nu\rho\tau}
```

With slashed momenta. Two of the slots of ε are now filled with momenta, $\varepsilon^{\mu\nu pq} = \varepsilon^{\mu\nu\rho\sigma} p_\rho q_\sigma$:

```
In [29]: dirac_trace(slash(p) * gamma(mu) * gamma5() * slash(q) * gamma(nu))
```

```
Out[29]: | 4i \epsilon^{\mu\nu pq}
```

The chiral projectors are just sums with γ^5 :

```
In [30]: PL()
```

Out[30]: $\left(\frac{1}{2} - \frac{1}{2}\gamma^5\right)$

and they behave as projectors, $P_L P_R = 0$ and $P_L P_L = P_L$:

In [31]: `contract(PL() * PR()), contract(PL() * PL())`

Out[31]: $\left(0, -\frac{1}{2}\gamma^5 + \frac{1}{2}\right)$

GAMMA5 IN D DIMENSIONS

In exactly four dimensions everything above is unique. In d dimensions nobody can define a γ^5 that anticommutes with all γ^μ and still gives the trace (5.5). Different “schemes” make different choices. feynsage does not choose for you without telling. It lets γ^5 anticommute (this is called NDR) only where the answer does not depend on the scheme. Two γ^5 cancel and one γ^5 with fewer than four other gamma matrices gives zero:

In [32]: `dirac_trace(PL() * gamma(mu) * PR() * gamma(nu), dim=d)`

Out[32]: $2g^{\mu\nu}$

Anything else is refused with a clear message:

In [33]: `try:
 dirac_trace(gamma(mu) * gamma(nu) * gamma(rho) * gamma(tau) * gamma5(), dim=d)
except DiracError as err:
 print(err)`

a trace with gamma5 and 4 other gamma matrices in d dimensions is ambiguous with an anticommuting gamma5 (NDR). Use dim=4, or do the gamma5 part in 4 dimensions.

8 Spinors

A spinor $u(p)$ is a column, so it stands at the right end of a line. $\bar{u}(p)$ is a row and stands at the left end. A line with a spinor at both ends is a number. feynsage draws it in square brackets.

In [34]: `J = ubar(p2, m) * gamma(mu) * u(p1, m)
J`

Out[34]: $[\bar{u}(p_2)\gamma^\mu u(p_1)]$

The complex conjugate turns $[\bar{u}(p_2)\gamma^\mu u(p_1)]^*$ into $[\bar{u}(p_1)\gamma^\mu u(p_2)]$. In general it reverses the matrices and takes $\gamma^0\Gamma^\dagger\gamma^0$ of the line. So γ^5 changes sign, i becomes $-i$ and P_L and P_R change places.

In [35]: `conjugate(J)`

Out[35]: $[\bar{u}(p_1)\gamma^\mu u(p_2)]$

`contract` knows the Dirac equation, $\not{p}u(p) = mu(p)$ and $\bar{u}(p)\not{p} = m\bar{u}(p)$. Here $\not{p}_2$ sits next to $\bar{u}(p_2)$ and $\not{p}_1$ next to $u(p_1)$, so both become m :

In [36]: `contract(ubar(p2, m) * slash(p2) * gamma(mu) * slash(p1) * u(p1, m))`

Out[36]: $m^2[\bar{u}(p_2)\gamma^\mu u(p_1)]$

For v spinors the sign is opposite, $\not{p}v(p) = -mv(p)$. Here $\not{p}_1$ must first jump over γ^5 , which gives one more sign:

In [37]: `contract(vbar(p2, m) * gamma(mu) * gamma5() * slash(p1) * v(p1, m))`

Out[37]: $-m[\bar{v}(p_2)\gamma^\mu\gamma^5v(p_1)]$

A product that makes no sense is stopped at once:

```
In [38]: try:
          gamma(mu) * ubar(p, m)
        except DiracError as err:
          print(err)
```

ubar(p, m) is a row spinor; it must stand at the left end of a line

9 Application 1: $e^+e^- \rightarrow \mu^+\mu^-$

This is the first process of Peskin and Schroeder chapter 5. An electron p_1 and a positron p_2 make a virtual photon, which makes a muon k_1 and an antimuon k_2 . We keep the electrons massless and the muon mass M_μ . Up to the factor e^2/q^2 the amplitude is

$$M = [\bar{v}(p_2)\gamma^\mu u(p_1)][\bar{u}(k_1)\gamma_\mu v(k_2)]$$

9.1 The amplitude and its square

```
In [39]: M_mu = var('M_mu')
M = vbar(p2) * gamma(mu) * u(p1) * ubar(k1, M_mu) * gamma(mu) * v(k2, M_mu)
M
```

Out[39]: $[\bar{v}(p_2)\gamma^\mu u(p_1)][\bar{u}(k_1)\gamma^\mu v(k_2)]$

Its conjugate. The summed index μ is renamed to μ_c , so that in MM^* no index appears four times:

```
In [40]: conjugate(M)
```

Out[40]: $[\bar{u}(p_1)\gamma^{\mu_c}v(p_2)][\bar{v}(k_2)\gamma^{\mu_c}u(k_1)]$

Now sum over all spins and divide by 4 for the average over the two initial spins. `spin_sum` puts $\sum u\bar{u} = \not{p} + m$ and $\sum v\bar{v} = \not{p} - m$ and lets FORM take the two traces.

```
In [41]: Msq = spin_sum(M * conjugate(M)) / 4
Msq
```

Out[41]: $8M_\mu^2(p_1 \cdot p_2) + 8(k_1 \cdot p_2)(k_2 \cdot p_1) + 8(k_1 \cdot p_1)(k_2 \cdot p_2)$

With $p = p_1$, $p' = p_2$, $k = k_1$ and $k' = k_2$ this is Peskin and Schroeder eq. (5.10) on p. 135, without its factor e^4/q^4 .

9.2 In the centre-of-mass frame

Each beam has energy E , the muons have $|\mathbf{k}| = \sqrt{E^2 - M_\mu^2}$ and leave at angle θ with $c = \cos\theta$. The scalar products are those listed on p. 136 of Peskin and Schroeder. We multiply by e^4/q^4 with $q^2 = 4E^2$:

```
In [42]: E, K, c, e = var('E K c e') # beam energy, |k|, cos(theta), charge
assume(E > M_mu, M_mu > 0)
frame = {dot(p1, p2): 2*E^2,
         dot(k1, p1): E^2 - E*K*c, dot(k2, p2): E^2 - E*K*c,
         dot(k2, p1): E^2 + E*K*c, dot(k1, p2): E^2 + E*K*c}
msq = (e^4 / (4*E^2)^2 * Msq).subs(frame).subs(K=sqrt(E^2 - M_mu^2))
msq.full_simplify()
```

Out[42]: $\frac{((E^2 - M_\mu^2)c^2 + E^2 + M_\mu^2)e^4}{E^2}$

This must be their eq. (5.11):

```
In [43]: peskin_511 = e^4 * ((1 + M_mu^2/E^2) + (1 - M_mu^2/E^2) * c^2)
         (msq - peskin_511).full_simplify()
```

```
Out[43]: 0
```

9.3 The cross section

For two particles into two particles in the centre-of-mass frame the cross section is $d\sigma/d\Omega = 1/(2E_{\text{cm}}^2) \cdot |\mathbf{k}|/(16\pi^2 E_{\text{cm}}) \cdot \frac{1}{4} \sum |M|^2$, the first line of Peskin and Schroeder eq. (5.12), with $E_{\text{cm}} = 2E$ and $e^2 = 4\pi\alpha$. Integrating over the angles gives the total cross section:

```
In [44]: a_em = var('a_em', latex_name=r'\alpha')
         Ecm = 2*E
         dsigma = 1/(2*Ecm^2) * sqrt(E^2 - M_mu^2)/(16*pi^2*Ecm) * msq.subs(e=sqrt(4*pi*a_em))
         sigma_tot = (2*pi * integrate(dsigma, c, -1, 1)).full_simplify()
         sigma_tot
```

```
Out[44]: (2*pi*E^2 + pi*M_mu^2)*sqrt(E^2 - M_mu^2)/6*E^5
```

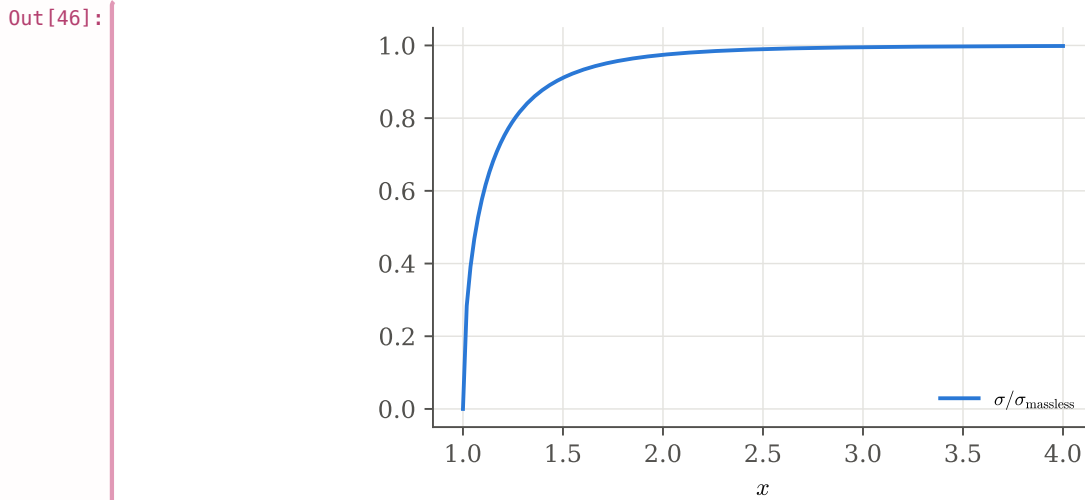
and it agrees with Peskin and Schroeder eq. (5.13):

```
In [45]: peskin_513 = 4*pi*a_em^2/(3*Ecm^2) * sqrt(1 - M_mu^2/E^2) * (1 + M_mu^2/(2*E^2))
         (sigma_tot - peskin_513).canonicalize_radical() # E > 0, so sqrt(E^2) = E
```

```
Out[45]: 0
```

Near the threshold $E = M_\mu$ the cross section starts from zero. Far above it, it goes to the massless value $4\pi\alpha^2/(3E_{\text{cm}}^2)$:

```
In [46]: from feynsage import plotting
         x = var('x') # x = E / M_mu
         ratio = (sigma_tot / (4*pi*a_em^2/(3*Ecm^2))).subs(E=x*M_mu).full_simplify()
         fig = plotting.quick_plot([ratio], (x, 1, 4), labels=[r'\sigma / \sigma_{\rm massless}'])
```



10 Application 2: Compton scattering

A photon k hits an electron p and leaves as k' . The electron leaves as p' . The photons now carry polarization vectors, $\varepsilon(k)$ for the one that comes in and $\varepsilon^*(k')$ for the one that goes out.

10.1 Polarization vectors

polarization makes $\varepsilon(k)$ and also its complex conjugate $\varepsilon^*(k)$. conjugate_vector gives the conjugate. latex_name only says how the symbols are printed, so that p' and k' look as in the book:

```
In [47]: kp, pp = momenta("kp pp", latex_name=["k'", "p'"])
e_in = polarization("e_in", k, latex_name=r"\epsilon") # photon in
e_out = polarization("e_out", kp, latex_name=r"\epsilon") # photon out
eo = conjugate_vector(e_out) # eps*(k')
```

```
Out[47]: |  $\epsilon'^*$ 
```

The two diagrams (s-channel and u-channel) of Compton scattering, without the factor $-ie^2$:

```
In [48]: Mc = (ubar(pp, m) * slash(eo) * (slash(p) + slash(k) + m) * slash(e_in) * u(p, m) /
(2*dot(p, k))
+ ubar(pp, m) * slash(e_in) * (slash(p) - slash(kp) + m) * slash(eo) * u(p, m) /
(-2*dot(p, kp)))
Mc
```

```
Out[48]: | 
$$\frac{1}{2(k \cdot p)} [\bar{u}(p') \not{\epsilon}'^* (\not{k} + \not{p} + m) \not{\epsilon} u(p)] - \frac{1}{2(k' \cdot p)} [\bar{u}(p') \not{\epsilon} (-\not{k}' + \not{p} + m) \not{\epsilon}'^* u(p)]$$

```

10.2 Spins and polarizations

polarization_sum replaces $\sum \varepsilon^\mu \varepsilon^{*\nu}$ by $-g^{\mu\nu}$. Then we put the particles on shell and use $k \cdot k' = p \cdot k - p \cdot k'$, which follows from $p + k = p' + k'$. The result has 15 terms:

```
In [49]: r = spin_sum(Mc * conjugate(Mc), rules={pp: p + k - kp})
r = polarization_sum(r, [e_in, e_out]) / 4
onshell = {dot(p, p): m^2, dot(k, k): 0, dot(kp, kp): 0, dot(k, kp): dot(p, k) - dot(p, kp)}
compton = r.subs(onshell)
len(compton.operands())
```

```
Out[49]: | 15
```

It must be Peskin and Schroeder eq. (5.87) on p. 162 with $e = 1$:

$$\frac{1}{4} \sum |M|^2 = 2e^4 \left[\frac{p \cdot k'}{p \cdot k} + \frac{p \cdot k}{p \cdot k'} + 2m^2 \left(\frac{1}{p \cdot k} - \frac{1}{p \cdot k'} \right) + m^4 \left(\frac{1}{p \cdot k} - \frac{1}{p \cdot k'} \right)^2 \right]$$

```
In [50]: pk, pkp = dot(p, k), dot(p, kp)
peskin_587 = 2*(pkp/pk + pk/pkp + 2*m^2*(1/pk - 1/pkp) + m^4*(1/pk - 1/pkp)^2)
(compton - peskin_587).simplify_rational()
```

```
Out[50]: | 0
```

THE WARD IDENTITY

Replacing $\sum \varepsilon^\mu \varepsilon^{*\nu}$ by $-g^{\mu\nu}$ is allowed because of the Ward identity. We can test it. With gauge="physical" feynsage sums only over the two real polarizations of each photon, with the help of a reference vector n . The answer must not change:

```
In [51]: r2 = polarization_sum(spin_sum(Mc * conjugate(Mc), rules={pp: p + k - kp}), [e_in,
e_out],
gauge="physical", n=p) / 4
(r2.subs(onshell) - compton).simplify_rational()
```

```
Out[51]: | 0
```

10.3 The Klein-Nishina formula

In the frame where the electron is at rest, $p \cdot k = m\omega$ and $p \cdot k' = m\omega'$ with $\omega' = \omega/(1 + (\omega/m)(1 - \cos\theta))$, Peskin and Schroeder eq. (5.89). With their cross-section formula on p. 163 we get $d\sigma/d\cos\theta$ and compare it with eq. (5.91), the Klein-Nishina formula:

```
In [52]: w = var('w', latex_name=r'\omega'); th = var('th', latex_name=r'\theta')
wp = w / (1 + w/m*(1 - cos(th))) # omega', Peskin (5.89)
lab = compton.subs({pk: m*w, pkp: m*wp}).subs(e=1)
dsig = 1/(2*w) * 1/(2*m) * 1/(8*pi) * wp^2/(w*m) * (4*pi*a_em)^2 * lab
peskin_591 = pi*a_em^2/m^2 * (wp/w)^2 * (wp/w + w/wp - sin(th)^2)
(dsig - peskin_591).full_simplify()
```

Out[52]: 0

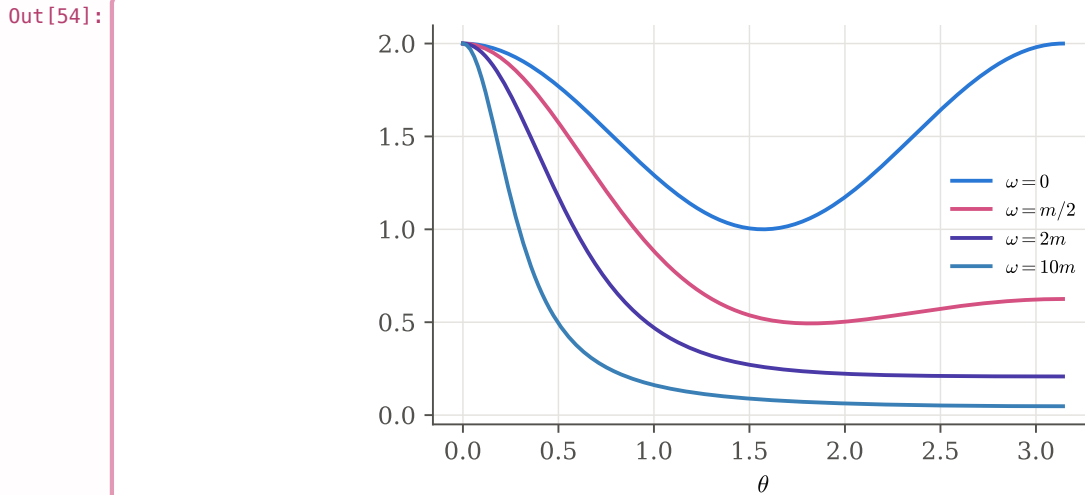
For small photon energy, $\omega \rightarrow 0$, it must become the classical Thomson result of eq. (5.92), $(\pi\alpha^2/m^2)(1 + \cos^2\theta)$ with total cross section $8\pi\alpha^2/(3m^2)$. The first number below is the difference from $(\pi\alpha^2/m^2)(1 + \cos^2\theta)$, the second the total:

```
In [53]: low = limit(peskin_591, w=0)
thomson = pi*a_em^2/m^2 * (1 + cos(th)^2)
(low - thomson).trig_simplify(), integrate(low * sin(th), th, 0, pi)
```

Out[53]: $\left(0, \frac{8\pi\alpha^2}{3m^2}\right)$

How the angular shape changes with energy. At low energy forward and backward scattering are equally likely. At high energy the photon mostly goes forward:

```
In [54]: th_ = var('theta')
curves = [1 + cos(th_)^2] + [(peskin_591 / (pi*a_em^2/m^2)).subs(w=W*m).subs(th=th_) for W
in (1/2, 2, 10)]
fig = plotting.quick_plot(curves, (th_, 0, pi),
labels=[r"$\omega = 0$", r"$\omega = m/2$", r"$\omega = 2m$", r"$\omega = 10m$"])
```



11 Application 3: left-handed currents

The weak interaction couples only to left-handed particles, through $\gamma^\mu P_L$. Take the current $[\bar{u}(p_2)\gamma^\mu P_L u(p_1)]$ of massless fermions. Its spin sum is a trace with two projectors:

```
In [55]: L = dirac_trace(slash(p2) * gamma(mu) * PL() * slash(p1) * gamma(nu) * PL())
L
```

Out[55]:
$$2p_1^\nu p_2^\mu + 2p_1^\mu p_2^\nu - 2(p_1 \cdot p_2)g^{\mu\nu} + 2i\epsilon^{\mu\nu\rho_1\rho_2}$$

The first three terms are half of what a normal vector current gives. The last term is new. It has $i\varepsilon$ and it changes sign under a mirror reflection, so it is the part that breaks parity.

Now multiply two such currents and contract, as in a weak decay where two left-handed currents meet:

```
In [56]: Lk = dirac_trace(slash(k2) * gamma(mu) * PL() * slash(k1) * gamma(nu) * PL())
          contract(L * Lk)
```

Out[56]:
$$16(k_1 \cdot p_1)(k_2 \cdot p_2)$$

Only one product of scalar products is left. Compare with one left-handed and one ordinary vector current (we take half of the vector trace so that the normal parts match):

```
In [57]: Lv = dirac_trace(slash(p2) * gamma(mu) * slash(p1) * gamma(nu)) / 2
          contract(Lv * Lk)
```

Out[57]:
$$8(k_1 \cdot p_2)(k_2 \cdot p_1) + 8(k_1 \cdot p_1)(k_2 \cdot p_2)$$

The ε terms of the two left-handed currents have added to the first product and cancelled the second. Squared amplitudes of weak decays have this shape, one product of two scalar products in place of a sum.

12 Application 4: colour

Quarks come in N colours ($N = 3$ in nature). A gluon has $N^2 - 1$ colours. The generators T^a are normalized with $\text{tr}(T^a T^b) = \delta^{ab}/2$. feynsage sums repeated colour indices with FORM and gives the answer as a function of N .

12.1 The basic colour factors

```
In [58]: a, b, cc, dd = color_indices("a b cc dd")
          i, j = quark_colors("i j")
          color_factor(T_color(a, i, j) * T_color(b, j, i))
```

Out[58]:
$$\frac{1}{2} \delta_{ab}$$

$(T^a T^a)_{ij} = C_F \delta_{ij}$ with $C_F = (N^2 - 1)/(2N)$, which is $4/3$ for $N = 3$:

```
In [59]: color_factor(color_chain(a, a, i=i, j=j))
```

Out[59]:
$$\frac{1}{2} N \delta_{ij} - \frac{\delta_{ij}}{2N}$$

$f^{acd} f^{bcd} = C_A \delta^{ab}$ with $C_A = N$:

```
In [60]: color_factor(f_color(a, cc, dd) * f_color(b, cc, dd))
```

Out[60]:
$$N \delta_{ab}$$

A harder one, $\text{tr}(T^a T^b T^a T^b) = -(N^2 - 1)/(4N)$, at $N = 3$:

```
In [61]: color_factor(color_trace(a, b, a, b), N=3)
```

Out[61]:
$$-\frac{2}{3}$$

12.2 The ratio R

In $e^+e^- \rightarrow q\bar{q}$ the photon makes a quark of colour i and an antiquark of colour j with amplitude proportional to δ_{ij} . Summing $|M|^2$ over the colours gives $\delta_{ij}\delta_{ji}$:

```
In [62]: color_factor(color_delta(i, j) * color_delta(j, i))
```

```
Out[62]: | N
```

So everything is as for muons, times N and times the squared quark charge Q_f^2 . The ratio $R = \sigma(e^+e^- \rightarrow \text{hadrons})/\sigma(e^+e^- \rightarrow \mu^+\mu^-)$ at energies well above the quark masses is therefore $N \sum_f Q_f^2$. With the five quarks u, d, s, c, b :

```
In [63]: Ncol = color_factor(color_delta(i, j) * color_delta(j, i), N=3)
charges = {'u': 2/3, 'd': -1/3, 's': -1/3, 'c': 2/3, 'b': -1/3}
R = Ncol * sum(Q^2 for Q in charges.values())
R
```

```
Out[63]: | 11/3
```

The colour factor of N is the reason this number is three times larger than it would be for colourless quarks.

12.3 Two gluons

In $q\bar{q} \rightarrow gg$ one diagram has the colour structure $(T^a T^b)_{ij}$. Its square summed over all colours is $\text{tr}(T^a T^b T^b T^a) = N C_F^2$:

```
In [64]: color_factor(color_chain(a, b, i=i, j=j) * color_chain(b, a, i=j, j=i))
```

```
Out[64]: | 1/4 N^3 - 1/2 N + 1/4 N
```

13 Application 5: a massive vector boson

This is problem 5.5 of Peskin and Schroeder (pp. 173 and 174). Add to QED a heavy photon B_μ of mass M_B that couples to electrons through $\Delta H = \int d^3x g \bar{\psi} \gamma^\mu \psi B_\mu$. The electron mass is set to zero everywhere. A massive vector has three polarizations $\varepsilon^{(i)}$ and their sum is

$$\sum_i \varepsilon_\mu^{(i)} \varepsilon_\nu^{(i)*} = -g_{\mu\nu} + \frac{k_\mu k_\nu}{M_B^2}$$

The book says that the second term can be dropped because B couples to a conserved current. We do not drop it. feynsage uses the full sum when the polarization is made with `mass=`, and we watch the $k_\mu k_\nu$ part go away by itself.

We compare every step with the worked solution of Xianyu [2], eqs. (5.80) to (5.93) on pp. 45 to 47.

13.1 Making a B and letting it decay

The electron p_1 and the positron p_2 annihilate into a B with momentum q :

```
In [65]: q = momenta("q") # the B
g, MB = var('g M_B')
eB = polarization("eB", q, mass=MB, latex_name=r"\varepsilon")
Ma = g * vbar(p2) * slash(conjugate_vector(eB)) * u(p1)
Ma
```

```
Out[65]: | g[\bar{v}(p_2) \not{\varepsilon} u(p_1)]
```

Sum over the spins and over the three polarizations of B . The terms with $1/M_B^2$ come from $k_\mu k_\nu/M_B^2$:

```
In [66]: ra = polarization_sum(spin_sum(Ma * conjugate(Ma)), eB)
ra
```

Out[66]:

$$8g^2(p_1 \cdot p_2) + \frac{8g^2(p_1 \cdot q)(p_2 \cdot q)}{M_B^2} - \frac{4g^2(p_1 \cdot p_2)q^2}{M_B^2}$$

Now put the particles on shell. Since $q = p_1 + p_2$, we have $q \cdot p_1 = q \cdot p_2 = p_1 \cdot p_2$ and $q^2 = 2p_1 \cdot p_2$. The $1/M_B^2$ terms cancel. Dividing by 4 averages over the electron and positron spins (Xianyu eq. 5.80), and dividing by 3 averages over the three polarizations of B for the decay (eq. 5.83):

In [67]:

```
ra = ra.subs({dot(p1, p1): 0, dot(p2, p2): 0, dot(q, p1): dot(p1, p2), dot(q, p2): dot(p1, p2),
              dot(q, q): 2*dot(p1, p2)})
ra / 4, ra / 3
```

Out[67]:

$$\left(2g^2(p_1 \cdot p_2), \frac{8}{3}g^2(p_1 \cdot p_2)\right)$$

The cross section is $\sigma = 1/(2s) \cdot |\overline{M}|^2 \cdot 2\pi\delta(s - M_B^2)$ and the decay rate is $\Gamma = 1/(2M_B) \cdot |\overline{M}|^2 \cdot 1/(8\pi)$ (two massless particles). Below are σ without its $\delta(s - M_B^2)$, then Γ , then the relation asked for in part (a), $\sigma - (12\pi^2/M_B)\Gamma$:

In [68]:

```
s = var('s')
sigma_B = 1/(2*s) * (ra/4).subs({dot(p1, p2): s/2}) * 2*pi # times delta(s - M_B^2)
Gamma_B = 1/(2*MB) * (ra/3).subs({dot(p1, p2): MB^2/2}) / (8*pi)
sigma_B, Gamma_B, sigma_B - 12*pi^2/MB * Gamma_B
```

Out[68]:

$$\left(\pi g^2, \frac{M_B g^2}{12\pi}, 0\right)$$

So $\sigma(e^+e^- \rightarrow B) = \pi g^2 \delta(s - M_B^2)$ and $\Gamma(B \rightarrow e^+e^-) = g^2 M_B / (12\pi)$, as in Xianyu eqs. (5.82) and (5.84). The relation of part (a) holds.

13.2 $e^+e^- \rightarrow \gamma + B$

Now the electron k_1 and the positron k_2 go to a photon p_1 and a B with momentum p_2 . There are two diagrams. In one the electron gives off the photon first, in the other it gives off the B first:

In [69]:

```
t = var('t'); uu = SR.var('uu', latex_name='u') # not var('u'): u is the spinor
ea = polarization("ea", p1, latex_name=r"\epsilon") # the photon
eb = polarization("eb", p2, mass=MB, latex_name=r"\varepsilon") # the B
ac, bc = conjugate_vector(ea), conjugate_vector(eb)
Mb = e*g*(vbar(k2) * slash(bc) * (slash(k1) - slash(p1)) * slash(ac) * u(k1) / (-2*dot(k1, p1))
          + vbar(k2) * slash(ac) * (slash(k1) - slash(p2)) * slash(bc) * u(k1) / (MB^2 - 2*dot(k1, p2)))
Mb
```

Out[69]:

$$-\frac{eg}{2(k_1 \cdot p_1)} [\bar{v}(k_2) \not{\epsilon} (\not{k}_1 - \not{p}_1) \not{\epsilon} u(k_1)] + \frac{eg}{M_B^2 - 2(k_1 \cdot p_2)} [\bar{v}(k_2) \not{\epsilon} (\not{k}_1 - \not{p}_2) \not{\epsilon} u(k_1)]$$

We square, sum over everything and divide by 4. The scalar products are written with $s = (k_1 + k_2)^2$, $t = (k_1 - p_1)^2$ and $u = (k_1 - p_2)^2$, and we use $s + t + u = M_B^2$:

In [70]:

```
mandelstam = {dot(k1, k1): 0, dot(k2, k2): 0, dot(p1, p1): 0, dot(p2, p2): MB^2,
              dot(k1, k2): s/2, dot(k1, p1): -t/2, dot(k2, p2): (MB^2 - t)/2,
              dot(k1, p2): (MB^2 - uu)/2, dot(k2, p1): -uu/2, dot(p1, p2): (s - MB^2)/2}
rb = polarization_sum(spin_sum(Mb * conjugate(Mb)), [ea, eb]) / 4
rb = rb.subs(mandelstam).subs(s=MB^2 - t - uu).simplify_rational()
rb
```

Out[70]:

$$\frac{2(2M_B^4 e^2 g^2 - 2M_B^2 e^2 g^2 t - 2M_B^2 e^2 g^2 u + e^2 g^2 t^2 + e^2 g^2 u^2)}{tu}$$

This must be Xianyu eq. (5.87):

$$\frac{1}{4} \sum |M|^2 = 2e^2 g^2 \left[\frac{u}{t} + \frac{t}{u} + \frac{2sM_B^2}{tu} \right]$$

```
In [71]: x587 = 2*e^2*g^2*(uu/t + t/uu + 2*s*MB^2/(t*uu))
         (rb - x587.subs(s=MB^2 - t - uu)).simplify_rational()
```

```
Out[71]: 0
```

WHY THE KK TERM COULD BE DROPPED

The book replaces the polarization sum of B by $-g_{\mu\nu}$ because k_μ dotted into the amplitude gives zero. Above we kept the full sum, so the answer does not depend on this. We can still test the claim. Put $\not{p}_2$ in place of $\not{\epsilon}^*$ of the B in the amplitude and square it. It vanishes:

```
In [72]: Mk = e*g*(vbar(k2) * slash(p2) * (slash(k1) - slash(p1)) * slash(ac) * u(k1) /
         (-2*dot(k1, p1))
         + vbar(k2) * slash(ac) * (slash(k1) - slash(p2)) * slash(p2) * u(k1) / (MB^2
         - 2*dot(k1, p2)))
         polarization_sum(spin_sum(Mk * conjugate(Mk)), ea).subs(mandelstam).subs(s=MB^2 - t -
         uu).simplify_rational()
```

```
Out[72]: 0
```

13.3 The angular distribution

In the centre-of-mass frame $t = (M_B^2 - s) \sin^2(\theta/2)$ and $u = (M_B^2 - s) \cos^2(\theta/2)$, where θ is the angle between the photon and the electron. With $d\sigma/d\Omega = 1/(2s) \cdot |\mathbf{p}_1|/(16\pi^2\sqrt{s}) \cdot |M|^2$ and $|\mathbf{p}_1| = (s - M_B^2)/(2\sqrt{s})$ we get $d\sigma/d\cos\theta$. Below is its difference from Xianyu eq. (5.91):

$$\frac{d\sigma}{d\cos\theta} = \frac{\alpha g^2 (1 - M_B^2/s)}{2s \sin^2\theta} \left[1 + \cos^2\theta + \frac{4sM_B^2}{(s - M_B^2)^2} \right]$$

```
In [73]: ph = var('ph') # th = theta, from the Compton part
         assume(s > MB^2, MB > 0)
         d0omega = 1/(2*s) * (s - MB^2)/(2*sqrt(s)) / (16*pi^2*sqrt(s)) * x587
         angles = {t: (MB^2 - s)*sin(th/2)^2, uu: (MB^2 - s)*cos(th/2)^2}
         dcos = 2*pi * d0omega.subs(angles).subs(e=sqrt(4*pi*a_em))
         x591 = a_em*g^2*(1 - MB^2/s)/(2*s*sin(th)^2) * (1 + cos(th)^2 + 4*s*MB^2/(s - MB^2)^2)
         (dcos - x591).subs(th=2*ph).expand_trig().simplify_full()
```

```
Out[73]: 0
```

For $M_B \rightarrow 0$ and $g \rightarrow e$ the B becomes a second photon. Then the formula must agree with $e^+e^- \rightarrow 2\gamma$ for massless electrons, which Peskin and Schroeder quote in problem 5.6(b) on p. 174 as $(2\pi\alpha^2/s)(1 + \cos^2\theta)/\sin^2\theta$:

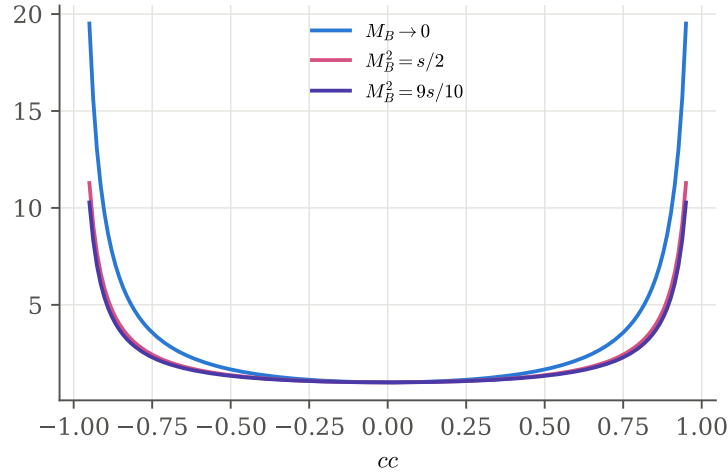
```
In [74]: (x591.subs({MB: 0, g: sqrt(4*pi*a_em)})) - 2*pi*a_em^2/s * (1 + cos(th)^2)/
         sin(th)^2).simplify_full()
```

```
Out[74]: 0
```

The shape for three masses of B , each curve divided by its value at $\theta = \pi/2$. All of them rise towards the beams like $1/\sin^2\theta$. For a light B the factor $1 + \cos^2\theta$ makes the rise a little steeper. For a heavy B the term $4sM_B^2/(s - M_B^2)^2$ is large and only $1/\sin^2\theta$ is left:

```
In [75]: cc = var('cc') # cos(theta)
         shape = (x591 / x591.subs(th=pi/2)).subs(th=arccos(cc))
         curves = [shape.subs({MB: sqrt(f*s)}).simplify_full() for f_ in (0, 1/2, 9/10)]
         fig = plotting.quick_plot(curves, (cc, -0.95, 0.95),
         labels=[r"$M_B \to 0$", r"$M_B^2 = s/2$", r"$M_B^2 = 9s/10$"])
```


Out[75]:



13.4 The photon along the beam

The cross section blows up at $\theta = 0$ and $\theta = \pi$. Near $\theta = 0$ we write $c = \cos \theta$. The cross section behaves as $A/(1-c)$ there, and A is

```
In [76]: c = var('c'); assume(c > -1, c < 1)
A = limit(((1 - c) * x591.subs(th=arccos(c))).simplify_full(), c=1)
A
```

Out[76]:

$$-\frac{M_B^4 \alpha g^2 + \alpha g^2 s^2}{2(M_B^2 s^2 - s^3)}$$

The electron mass cuts the angle off at $\theta^2 \sim m_e^2/s$, so the integral of $A/(1-c)$ up to there gives $A \log(s/m_e^2)$ to leading logarithm (Xianyu eq. 5.92). Part (c) of the problem says that this must be the same as making a B from an electron that has given a fraction x of its energy to a photon, with the Weizsäcker-Williams function

$$f(x) = \frac{\alpha}{2\pi} \frac{1 + (1-x)^2}{x} \log\left(\frac{s}{m_e^2}\right)$$

times $\sigma(e^+e^- \rightarrow B) = \pi g^2 \delta(M_B^2 - (1-x)s)$. The delta function sets $x = 1 - M_B^2/s$ and gives a factor $1/s$. The difference from A is

```
In [77]: x = var('x')
f = a_em/(2*pi) * (1 + (1 - x)^2)/x
ww = (f * pi*g^2/s).subs(x=1 - MB^2/s)
integral done
(A - ww).simplify_full()
```

f(x) without its log(s/m_e^2)
pi g^2 delta(M_B^2 - (1 - x) s), x

Out[77]:

0

The two agree (Xianyu eq. 5.93). The photon that goes along the beam is just the electron losing some energy before it makes the B .

14 Application 6: the equivalent photon approximation

This is problem 6.2 of Peskin and Schroeder (pp. 209 and 210). A very fast electron with energy E scatters off a target and leaves with energy E' at a small angle θ . The photon between them has $q = p - p'$ and $q^2 \approx -2EE'(1 - \cos \theta)$, which goes to zero in the forward direction. The question is in what sense this photon becomes a real one. We follow the steps of the problem and compare with Xianyu's solution [2], eqs. (6.12) to (6.25) on pp. 53 to 55.

The amplitude is $\mathcal{M} = (-ie)\bar{u}(p')\gamma^\mu u(p)(-ig_{\mu\nu}/q^2)\widehat{\mathcal{M}}^\nu(q)$, where $\widehat{\mathcal{M}}^\nu$ is the (unknown) coupling of the photon to the target. The problem writes the electron current as

$$\bar{u}(p')\gamma^\mu u(p) = Aq^\mu + B\tilde{q}^\mu + C\varepsilon_1^\mu + D\varepsilon_2^\mu$$

with $\tilde{q} = (q^0, -\mathbf{q})$ and two unit vectors ε_1 (in the plane of scattering) and ε_2 (out of it), both at right angles to $\mathbf{q}$.

14.1 C and D from traces

We need C and D for an electron of definite helicity. For a massless spinor $u_R(p)\bar{u}_R(p) = P_R\not{p}$, so the products of C and D are traces with one projector. feynsage keeps ε_1 and ε_2 as vectors, like momenta:

```
In [78]: e1, e2 = momenta("e1 e2", latex_name=[r"\epsilon_1", r"\epsilon_2"])
CC = dirac_trace(slash(pp) * slash(e1) * PR() * slash(p) * slash(e1)) # |C|^2
DD = dirac_trace(slash(pp) * slash(e2) * PR() * slash(p) * slash(e2)) # |D|^2
CD = dirac_trace(slash(pp) * slash(e1) * PR() * slash(p) * slash(e2)) # C D^*
```

```
Out[78]: 2(\epsilon_1 \cdot p')(\epsilon_2 \cdot p) + 2(\epsilon_1 \cdot p)(\epsilon_2 \cdot p') - 2(\epsilon_1 \cdot \epsilon_2)(p \cdot p') - 2i\epsilon^{\epsilon_1\epsilon_2}pp'
```

Now put in the frame of the problem. The electron comes in along z , $p = (E, 0, 0, E)$, and leaves in the xz plane. The Levi-Civita symbol becomes a determinant of the components ($\epsilon^{0123} = +1$). We keep the leading order in θ . The three numbers are $|C|^2$, $|D|^2$ and CD^* for the right-handed electron:

```
In [79]: Ep = SR.var('Ep', latex_name="E'")
qt = momenta("qt", latex_name=r"\tilde{q}")
assume(E > Ep, Ep > 0, th > 0, th < pi)
vec = {p: vector([E, 0, 0, E]), pp: vector([Ep, Ep*sin(th), 0, Ep*cos(th)]),
       e1: vector([0, Ep*cos(th) - E, 0, -Ep*sin(th)]) / sqrt(E^2 + Ep^2 - 2*E*Ep*cos(th)),
       e2: vector([0, 0, 1, 0]),
       qt: vector([E - Ep, Ep*sin(th), 0, Ep*cos(th) - E])} # q~ = (q0, -q)
mdot = lambda a, b: a[0]*b[0] - a[1]*b[1] - a[2]*b[2] - a[3]*b[3]
frame = {dot(a, b): mdot(vec[a], vec[b]) for a in vec for b in vec}
frame[epsilon(e1, e2, p, pp)] = -matrix([vec[e1], vec[e2], vec[p], vec[pp]]).det() #
eps^{0123} = +1
w0 = SR.wild(0)
root = (E^2 - 2*E*Ep + Ep^2)^w0 == (E - Ep)^(2*w0) # sqrt((E - E')^2) = E - E' because
E > E'
small = lambda X: X.subs(frame).series(th,
3).truncate().simplify_full().subs(root).factor()
[small(X) for X in (CC, DD, CD)]
```

```
Out[79]: \left[ \frac{(E + E')^2 E E' \theta^2}{(E - E')^2}, E E' \theta^2, \frac{E(i E + i E') E' \theta^2}{E - E'} \right]
```

Xianyu's eq. (6.19) gives $C_\pm = -\sqrt{EE'}(E + E')/|E - E'|\theta$ and $D_\pm = \pm i\sqrt{EE'}\theta$. The differences are zero:

```
In [80]: C2 = E*Ep*(E + Ep)^2/(E - Ep)^2 * th^2
D2 = E*Ep * th^2
CDs = I*E*Ep*(E + Ep)/(E - Ep) * th^2
[(small(X) - Y).simplify_full() for X, Y in ((CC, C2), (DD, D2), (CD, CDs))]
```

```
Out[80]: [0, 0, 0]
```

For the left-handed electron CD^* has the opposite sign, so in the average over the two helicities the mixed term goes away:

```
In [81]: CDL = dirac_trace(slash(pp) * slash(e1) * PL() * slash(p) * slash(e2))
(CD + CDL).subs(frame).simplify_full()
```

```
Out[81]: 0
```

In a separate script (not shown) we also checked (6.16) and (6.17) themselves, with explicit spinors in the chiral basis of Peskin and Schroeder and the spinors of Xianyu's eq. (6.15). They agree, including the sign of C and the factor i of D .

A TRAP IN SAGE: SQUARE ROOTS

The small-angle form has $\sqrt{(E - E')^2}$. Sage's `canonicalize_radical()` replaces it by $-(E - E')$ even after `assume(E > Ep)`. That flips the sign of CD^* and of nothing else, so it is easy to miss. Above we tell Sage $\sqrt{(E - E')^2} = E - E'$ ourselves, with a pattern (SR.wild) that also catches the power $-1/2$.

14.2 Part (a): B is small

The current is conserved, $q \cdot J = 0$, so $Aq^2 + Bq \cdot \tilde{q} = 0$. Dotting J with q and $\tilde{q}$ gives $B = q^2(\tilde{q} \cdot J)/(q^4 - (q \cdot \tilde{q})^2)$. Below are $\sum |q \cdot J|^2$ and $|B|^2$ averaged over spins:

```
In [82]: J = ubar(pp) * (slash(p) - slash(pp)) * u(p)          # q.J with q = p - p'
          Jt = ubar(pp) * slash(qt) * u(p)                    # q~.J
          Q = vec[p] - vec[pp]
          q2_, qqt = mdot(Q, Q), mdot(Q, vec[qt])
          B2 = (spin_sum(Jt * conjugate(Jt)) / 2).subs(frame) * q2_^2/(q2_^2 - qqt^2)^2
          spin_sum(J * conjugate(J)).subs(frame).simplify_full(), B2.series(th,
          5).truncate().simplify_full()
```

```
Out[82]: 
$$\left(0, \frac{E^3 E'^3 \theta^4}{E^6 - 6 E^5 E' + 15 E^4 E'^2 - 20 E^3 E'^3 + 15 E^2 E'^4 - 6 E E'^5 + E'^6}\right)$$

```

So B is of order θ^2 , as the problem says.

14.3 Parts (c) to (e): the photon spectrum

With the average over helicities, $1/2 \sum |\mathcal{M}|^2 = e^2/(q^2)^2 EE' \theta^2 [(E + E')/(E - E')]^2 |\varepsilon_1 \cdot \widehat{\mathcal{M}}|^2 + |\varepsilon_2 \cdot \widehat{\mathcal{M}}|^2]$, which is Xianyu eq. (6.21). Part (e) assumes that the target does not care about the photon polarization, so $|\varepsilon_1 \cdot \widehat{\mathcal{M}}|^2 = |\varepsilon_2 \cdot \widehat{\mathcal{M}}|^2 = |\widehat{\mathcal{M}}|^2/2$, where $|\widehat{\mathcal{M}}|^2$ is summed over the two polarizations. Writing $E' = (1 - x)E$:

```
In [83]: Mh = SR.var('Mh', latex_name=r'\widehat{\mathcal{M}}|^2')
          avg = (C2 * Mh/2 + D2 * Mh/2).subs(Ep=(1 - x)*E)
          avg.factor()
```

```
Out[83]: 
$$-\frac{(x^2 - 2x + 2)E^2 |\widehat{\mathcal{M}}|^2 \theta^2 (x - 1)}{x^2}$$

```

Part (d) keeps the electron mass in $q^2 = -2(EE' - |p| |p'| \cos \theta) + 2m^2$. For small θ and m (scaled together) Sage gives the expansion, and its difference from $-(1 - x)E^2 \theta^2 - x^2 m^2/(1 - x)$, the form in Xianyu's part (d):

```
In [84]: lt = var('lt')          # theta and m both small: scale them together
          Epx = (1 - x)*E
          q2m = -2*(E*Epx - sqrt(E^2 - m^2)*sqrt(Epx^2 - m^2)*cos(th)) + 2*m^2
          assume(x > 0, x < 1)
          q2_small = q2m.subs(th=lt*th, m=lt*m).series(lt, 3).truncate().subs(lt=1)
          q2_small = q2_small.subs((E^2*x^2 - 2*E^2*x + E^2)^w0 == (E*(1 - x))^(2*w0)).subs((E^2)^w0
          == E^(2*w0))
          q2_small.simplify_full(), (q2_small - (-(1 - x)*E^2*th^2 - x^2*m^2/(1 -
          x))).simplify_full()
```

```
Out[84]: 
$$\left(-\frac{2 E^2 \theta^2 x - E^2 \theta^2 - (E^2 \theta^2 + m^2) x^2}{x - 1}, 0\right)$$

```

So $q^2 = -(1 - x)E^2(\theta^2 + a^2)$ with $a^2 = x^2 m^2/((1 - x)^2 E^2)$. The mass now stops the forward divergence. The angular integral is

```
In [85]: a2, tmax = var('a2 tmax'); assume(a2 > 0, tmax > 0)
          I_th = integrate(th^3/(th^2 + a2)^2, th, 0, tmax)
          I_th, (I_th + log(a2)/2 - log(tmax)).subs(a2=0).simplify_log()
```

Out[85]:
$$\left(\frac{tmax^2 \log(tmax^2 + a_2)}{2(tmax^2 + a_2)} + \frac{a_2 \log(tmax^2 + a_2)}{2(tmax^2 + a_2)} + \frac{a_2}{2(tmax^2 + a_2)} - \frac{1}{2} \log(a_2) - \frac{1}{2}, -\frac{1}{2} \right)$$

that is $1/2 \log(\theta_{\max}^2/a^2) - 1/2$ for small a . To leading logarithm it is $1/2 \log(E^2/m^2)$, a positive number.

Now we put everything together. The phase space of the electron is $d^3p'/((2\pi)^3 2E') = E' dE' \theta d\theta/(8\pi^2)$ (after the trivial angle), with $dE' = E dx$. The flux is $1/(2E \cdot 2M_t)$ for a target of mass M_t at rest. A real photon of energy $x E$ hitting the same target has $\sigma_\gamma = 1/(2xE \cdot 2M_t) \cdot |\widehat{\mathcal{M}}|^2/2$ times the same target phase space. Then $d\sigma = dx N_\gamma(x) \sigma_\gamma(xE)$ with

```
In [86]: Mt, L = var('M_t L') # target mass; L = log(E^2/m^2)
q2_0 = -(1 - x)*E^2*(th^2 + a2)
per = (Ep*x*E * th/(8*pi^2) * e^2/q2_0^2 * avg).simplify_full() # times dx dtheta and the
target phase space
sigma_gamma = (Mh/2) / (2*x*E * 2*Mt) # photon cross section, per unit of target
phase space
N = (per / th^3 * (th^2 + a2)^2 * L/2) / (2*E * 2*Mt) / sigma_gamma
N.subs(e=sqrt(4*pi*a_em)).simplify_full()
```

Out[86]:
$$\frac{L\alpha x^2 - 2L\alpha x + 2L\alpha}{2\pi x}$$

where $L = \log(E^2/m^2)$. This is the Weizsäcker-Williams distribution of Peskin and Schroeder (p. 210),

$$N_\gamma(x) = \frac{\alpha}{2\pi} \frac{1 + (1-x)^2}{x} \log\left(\frac{s}{m^2}\right)$$

to leading logarithm. It is the same $f(x)$ that appeared in Application 5.

NOTES ON THE WORKED SOLUTION

The final result is right, but a few steps on pp. 54 and 55 of [2] do not follow from each other. (1) The second line of eq. (6.22) has a factor $1/3$ (an average over three polarizations), and the third line is exactly three times the second. With the assumption of part (e) the factor is $1/2$ per polarization, as above. (2) Eq. (6.23) has $(1 - \cos\theta)^4$ in the denominator. It must be $(1 - \cos\theta)^2$, as in (6.22), to give $\int d\theta/\theta$. (3) Eq. (6.24) says $-1/2 \log(E^2/m^2)$, but the integrand is positive and the integral is $+1/2 \log(E^2/m^2)$, as computed above. The overall sign of (6.25) still comes out right, because the minus sign in (6.22) comes from $dE' = -E dx$ and goes away when the limits of x are put in order. (4) Eq. (6.25) gives N_γ only if $|\widehat{\mathcal{M}}_\mu|^2$ there means the square for one polarization, which is the average over the two.

15 Application 7: new particles and $g - 2$

This is problem 6.3 of Peskin and Schroeder (p. 210). Any new particle that couples to the electron changes the vertex $\Gamma^\mu = \gamma^\mu F_1(q^2) + i\sigma^{\mu\nu} q_\nu/(2m) F_2(q^2)$ at one loop and so changes $a = (g - 2)/2 = F_2(0)$. We take a scalar h (a Higgs boson) with $H_{\text{int}} = \int d^3x (\lambda/\sqrt{2}) h \bar{\psi} \psi$ and a pseudoscalar a (an axion) with $H_{\text{int}} = \int d^3x (i\lambda/\sqrt{2}) a \bar{\psi} \gamma^5 \psi$. We compare with Xianyu [2], eqs. (6.26) to (6.40) on pp. 55 to 57.

15.1 The Higgs loop in one call

The beginner notebook computes the QED $F_2(0) = \alpha/(2\pi)$ with `projector` and `form_factor`. Here the photon in the loop is replaced by the Higgs, so the line along the electron is $(\not{p}_2 - \not{l} + m)\gamma^\mu(\not{p}_1 - \not{l} + m)$ with no gamma matrix at the two Higgs vertices. The Higgs carries the loop momentum l :

```
In [87]: from feynsage.dirac import projector, form_factor
q2 = SR.var('q2', latex_name='q^2'); mh = SR.var('mh', latex_name='m_h'); lam =
SR.var('lam', latex_name=r'\lambda')
P = projector('F2', 'mu', 'p1', 'm', 'p2', 'm')
kin = {'p1^2': 'm^2', 'p2^2': 'm^2', 'p1.p2': 'm^2 - q2/2'}
higgs = form_factor(['p2 - l + m', 'mu', 'p1 - l + m'], P, ['l', 'mh'], ['l - p2', 'm'],
['l - p1', 'm'], kin=kin)
Lh = higgs.coefficients()['1']
uv_part(Lh).simplify_full()
```

```
Out[87]: 0
```

The $1/\varepsilon$ pole of F_2 is zero, as it must be. The factors in front are $(-i\lambda/\sqrt{2})^2$ for the two vertices, i^3 for the three propagators and $i/(16\pi^2)$, because `loop` is normalized as $\int d^4l/(\pi^2)$ while the Feynman rules have $\int d^4l/(2\pi)^4$. Together they give $-\lambda^2/(32\pi^2)$. At $q^2 = 0$:

```
In [88]: F2h = -lam^2/(32*pi^2) * finite_part(Lh, 1).subs(q2=0)
from feynsage import tex
LatexExpr(r"\begin{aligned} \delta F_2(0) &= -\frac{\lambda^2}{32\pi^2}\Big[ "
+ tex.lines(finite_part(Lh, 1).subs(q2=0).expand(), per_line=3) + r" \Big] "
+ tex.lines(finite_part(Lh, 1).subs(q2=0).expand(), per_line=3) + r" \Big] "
+ tex.lines(finite_part(Lh, 1).subs(q2=0).expand(), per_line=3) + r" \Big] "
\end{aligned}")
```

```
Out[88]:
```

$$\begin{aligned} \delta F_2(0) = & -\frac{\lambda^2}{32\pi^2} \left[3m_h^2 C_0(m^2, m^2, 0, m, m_h, m) - \frac{3m_h^4 C_0(m^2, m^2, 0, m, m_h, m)}{4m^2} - \frac{3m_h^2 \Lambda(0; m, m)}{4m^2} \right. \\ & + \frac{5m_h^2 \Lambda(m^2; m, m_h)}{4m^2} - \frac{9m_h^2 \log\left(\frac{m^2}{m_h^2}\right)}{4m^2} + \frac{5m_h^4 \log\left(\frac{m^2}{m_h^2}\right)}{8m^4} \\ & - \frac{3m_h^2 \log\left(\frac{1}{m^2}\right)}{4m^2} + \frac{3m_h^2 \log\left(\frac{1}{m_h^2}\right)}{4m^2} + \frac{m_h^2}{2m^2} \\ & + \frac{3}{2} \Lambda(0; m, m) - 2\Lambda(m^2; m, m_h) + 2\log\left(\frac{m^2}{m_h^2}\right) \\ & \left. + 2\log\left(\frac{1}{m^2}\right) - 2\log\left(\frac{1}{m_h^2}\right) \right] \end{aligned}$$

Xianyu's eq. (6.34) has the same thing as one Feynman-parameter integral,

$$\delta F_2(0) = \frac{\lambda^2}{(4\pi)^2} \int_0^1 dx \frac{(1-x)^2(1+x)}{(1-x)^2 + x(m_h/m)^2}$$

The table has $(m_h/m)^2$, then feynsage, then eq. (6.34), each to 20 digits:

```
In [89]: x = var('x')
matrix(SR, [[R, F2h.subs(m=1, mh=sqrt(R), lam=1).n(digits=60).n(digits=20),
(integrate((1-x)^2*(1+x)/((1-x)^2 + x*R), x, 0, 1)/
(4*pi^2).n(digits=60).n(digits=20))
for R in (1/2, 10, 1000]]) # R = (m_h/m)^2, feynsage, eq.
(6.34)
```

```
Out[89]:
```

$$\begin{pmatrix} \frac{1}{2} & 0.0040101326337403856962 & 0.0040101326337403856962 \\ 10 & 0.0010484312114206570667 & 0.0010484312114206570667 \\ 1000 & 0.000036466850174497883945 & 0.000036466850174497883945 \end{pmatrix}$$

For a heavy Higgs, eq. (6.35) says $\delta F_2(0) \approx \lambda^2/((4\pi)^2 R)[\log R - 7/6]$ with $R = (m_h/m)^2$. We write $R = 2 + a + 1/a$, which factorizes the denominator, and expand in $\omega = 1/a$. What is left after the approximation is smaller by one more power of $1/R$:

```
In [90]: a, w = var('a w'); assume(a > 1, w > 0)
Ih = integrate(((1-x)^2*(1+x)/((x+a)*(x+1/a))).partial_fraction(x), x, 0, 1)
R_ = 2 + a + 1/a # (1-x)^2 + x R_ = (x+a)(x+1/a)
(Ih - (log(R_) - 7/6)/R_).subs(a=1/w).series(w, 3).truncate().simplify_full()
```

Out [90]:

$$-3\omega^2 \log(\omega) - \frac{13}{4}\omega^2$$

NUMBERS AT A LARGE MASS RATIO

At $q^2 = 0$ with $p_1^2 = p_2^2 = m^2$ the triangle C_0 sits where its Gram determinant vanishes. While writing this section we found that feynsage computed C_0 there with a double-precision fallback, which was 10% wrong at $m_h/m = 1000$. It now uses the closed form next to that point and extrapolates, good to about 50 digits in this case. The formula for F_2 still has terms of size $(m_h/m)^4$ that cancel, so ask for many digits, for example `.n(digits=60)`, when m_h/m is large.

15.2 Part (b): the numbers

For the electron, $\lambda = 3 \times 10^{-6}$, $m = 0.511$ MeV and $m_h = 60$ GeV. For the muon, $\lambda = 6 \times 10^{-4}$ and $m = 105.66$ MeV, with $m_h = 60$ GeV and with a very light Higgs, $m_h = m_\mu/1000$:

```
In [91]: electron = F2h.subs(lam=3/10^6, m=511/1000, mh=60000) # masses in MeV
muon_60 = F2h.subs(lam=6/10^4, m=10566/100, mh=60000)
muon_light = F2h.subs(lam=6/10^4, m=10566/100, mh=10566/100000) # m_h = m_mu/1000
[v.n(digits=60).n(digits=4) for v in (electron, muon_60, muon_light)]
```

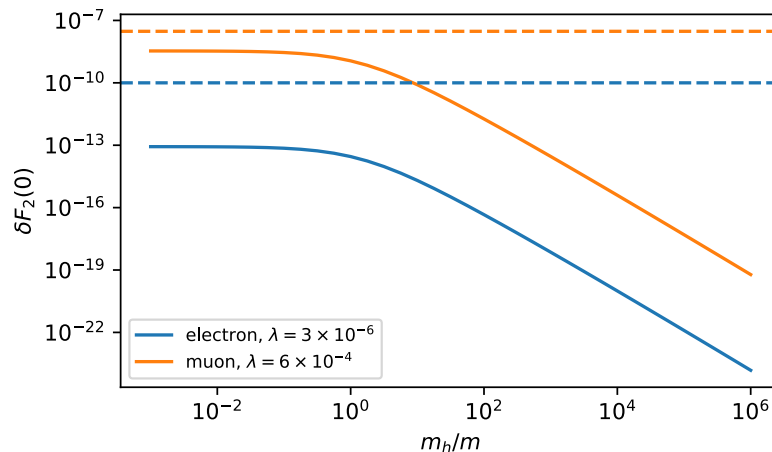
Out [91]:

$$[9.169 \times 10^{-23}, 8.142 \times 10^{-14}, 3.412 \times 10^{-9}]$$

The electron value is 9×10^{-23} , far below the bound 10^{-10} . For the muon even a Higgs of almost zero mass gives only 3.4×10^{-9} , below 3×10^{-8} . The reason is the limit $m_h \rightarrow 0$ of the integral above, $\lambda^2/(4\pi)^2 \cdot 3/2$. So with these couplings the muon $g - 2$ gives no limit on m_h at all. The plot shows $\delta F_2(0)$ against m_h/m , with the two bounds as dashed lines:

```
In [92]: import numpy as np, matplotlib.pyplot as plt
ratios = np.logspace(-3, 6, 37) # m_h / m
unit = [float(F2h.subs(lam=1, m=1, mh=QQ(float(rv))).n(digits=60).real()) for rv in ratios]
fig, ax = plt.subplots(figsize=(5.2, 3.2))
ax.loglog(ratios, [3e-6**2*u_ for u_ in unit], label=r"electron, $\lambda = 3 \times 10^{-6}$")
ax.loglog(ratios, [6e-4**2*u_ for u_ in unit], label=r"muon, $\lambda = 6 \times 10^{-4}$")
ax.axhline(1e-10, ls="--", color="C0"); ax.axhline(3e-8, ls="--", color="C1")
ax.set_xlabel(r"$m_h/m$"); ax.set_ylabel(r"$\delta F_2(0)$"); ax.legend(fontsize=8)
fig.tight_layout()
```

Out [92]:



15.3 Part (c): the axion

Each axion vertex is $-i \cdot i\lambda/\sqrt{2}\gamma^5 = (\lambda/\sqrt{2})\gamma^5$, so the line gets a γ^5 at each end and the factor in front is $+\lambda^2/(32\pi^2)$:

```
In [93]: ma = SR.var('ma', latex_name='m_a')
axion = form_factor(['g5', 'p2 - l + m', 'mu', 'p1 - l + m', 'g5'], P, ['l', 'ma'], ['l - p2', 'm'], ['l - p1', 'm'], kin=kin)
F2a = lam^2/(32*pi^2) * finite_part(axion.coefficients()['1'], 1).subs(q2=0)
matrix(SR, [[R, F2a.subs(m=1, ma=sqrt(R), lam=1).n(digits=60).n(digits=20),
(-integrate((1 - x)^3/((1 - x)^2 + x*R), x, 0, 1)/
(4*pi)^2).n(digits=60).n(digits=20)]
for R in (1/100, 2, 1000)]) # R = (m_a/m)^2, feynsage, eq.
(6.40)
```

```
Out[93]:
```

$$\begin{pmatrix} \frac{1}{100} & -0.0030708424399288822139 & -0.0030708424399288822139 \\ 2 & -0.0014948452025791385270 & -0.0014948452025791385270 \\ 1000 & -0.000032306640212805407293 & -0.000032306640212805407293 \end{pmatrix}$$

The columns are $(m_a/m)^2$, feynsage and Xianyu's eq. (6.40) before its approximation, $-\lambda^2/(4\pi)^2 \int_0^1 dx(1-x)^3/((1-x)^2 + x(m_a/m)^2)$. The axion makes a smaller, the Higgs makes it larger.

The problem says that the axion may be as light as the electron or lighter, and the solution treats only a heavy axion. Both limits:

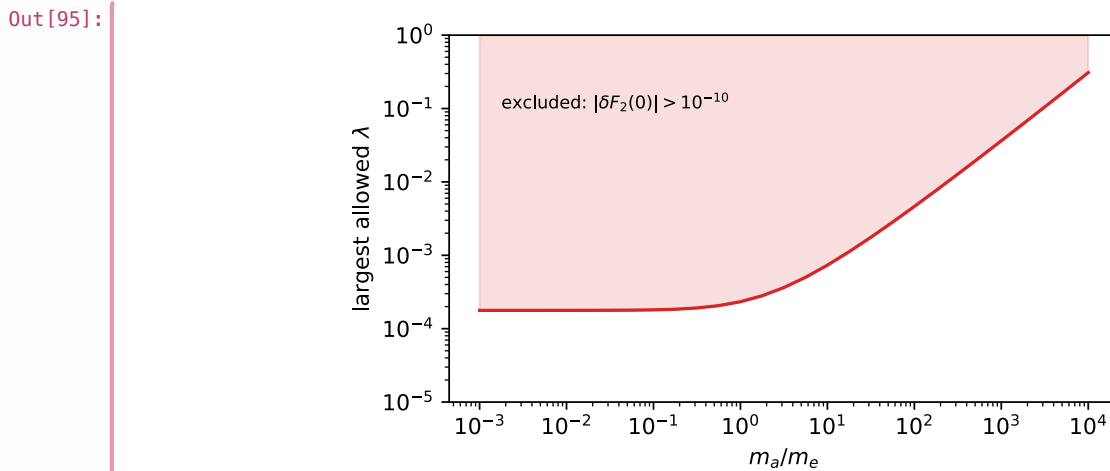
```
In [94]: Ia = integrate(((1 - x)^3/((x + a)*(x + 1/a))).partial_fraction(x), x, 0, 1)
light = -lam^2/(4*pi)^2 * integrate((1 - x)^3/(1 - x)^2, x, 0, 1) # m_a -> 0
heavy = (Ia - (log(R_) - 11/6)/R_).subs(a=1/w).series(w, 3).truncate().simplify_full()
light, heavy, sqrt(10^-10 / (light/lam^2).abs()).n(digits=3)
```

```
Out[94]:
```

$$\left(-\frac{\lambda^2}{32\pi^2}, -5\omega^2 \log(\omega) - \frac{89}{12}\omega^2, 0.000178 \right)$$

For $m_a \rightarrow 0$ the shift is $-\lambda^2/(32\pi^2)$, so $|\delta F_2(0)| < 10^{-10}$ needs $\lambda < 1.8 \times 10^{-4}$. For a heavy axion the approximation of eq. (6.40), $-\lambda^2/((4\pi)^2 R)[\log R - 11/6]$, is right up to terms of order $1/R^2$. The excluded region in the plane of m_a and λ :

```
In [95]: ratios = np.logspace(-3, 4, 29) # m_a / m
lam_max = [float(sqrt(1e-10 / abs(F2a.subs(lam=1, m=1,
ma=QQ(float(rv))).n(digits=60).real())))) for rv in ratios]
fig, ax = plt.subplots(figsize=(5.2, 3.2))
ax.loglog(ratios, lam_max, color="C3")
ax.fill_between(ratios, lam_max, 1, color="C3", alpha=0.15)
ax.text(ratios[1], 0.1, "excluded:  $|\delta F_2(0)| > 10^{-10}$ ", fontsize=8)
ax.set_xlabel(r"$m_a/m_e$"); ax.set_ylabel(r"largest allowed  $\lambda$ "); ax.set_ylim(1e-5, 1)
fig.tight_layout()
```



Below $m_a \approx m_e$ the limit is flat at $\lambda \approx 1.8 \times 10^{-4}$. Above it the limit grows roughly like m_a/m_e , the “ $\lambda m/m_a \gtrsim 10^{-5}$ is excluded” of the solution, with a logarithm.

16 Conventions: the book or the note, Minkowski or Euclidean

Two conventions are built in. "standard" follows Peskin and Schroeder and calls the dimension d . "note" follows my lecture note on Feynman integrals, which writes $D = 4 - 2\varepsilon$:

```
In [96]: set_convention("note")
         spacetime_dimension()
```

```
Out[96]: |
```

$$D$$

```
In [97]: set_convention("standard")
         spacetime_dimension()
```

```
Out[97]: |
```

$$d$$

Both use the metric $(+, -, -, -)$. For loop integrals one often rotates to Euclidean space. With `euclidean=True` the algebra is $\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}$ with $\gamma^5 = \gamma_1\gamma_2\gamma_3\gamma_4$ and $\varepsilon_{1234} = +1$, so the sign and the i of the γ^5 trace change:

```
In [98]: set_convention(euclidean=True)
         t5 = dirac_trace(gamma(mu) * gamma(nu) * gamma(rho) * gamma(tau) * gamma5())
         set_convention(euclidean=False)
         t5
```

```
Out[98]: |
```

$$4\epsilon^{\mu\nu\rho\tau}$$

A scalar result moves between the two with the rule of the note, $p \cdot q = -p_E \cdot q_E$, so that $s = p^2 = -p_E^2$:

```
In [99]: to_euclidean(dot(p, p) - m^2)
```

```
Out[99]: |
```

$$-m^2 - p^2$$

17 Speed and what happens inside

feynsage starts one FORM process the first time it needs one and keeps it running. A small trace then costs about a tenth of a millisecond:

```
In
[100]: import time
       exprs = [slash(p) * gamma(mu) * slash(q) * gamma(nu) * n for n in range(1, 201)]
       t0 = time.time(); one_by_one = [dirac_trace(x) for x in exprs]; t1 = time.time()
       together = dirac_trace(exprs); t2 = time.time()
       print("200 traces one by one: %.3f s, in one FORM run: %.3f s" % (t1 - t0, t2 - t1))
```

```
| 200 traces one by one: 0.025 s, in one FORM run: 0.019 s
```

To see what FORM was asked, add `debug=True`. You see the FORM program feynsage wrote and the answer FORM printed:

```
In
[101]: dirac_trace(slash(p) * gamma(mu) * slash(q) * gamma(mu), debug=True)
```

```
--- FORM program ---
Vectors p,q;
Indices mu;
Off statistics;
Local FSE1 = g_(1,p,mu,q,mu);
trace4,1;
contract;
Format nospaces;
Print;
.end
```



```
--- FORM output ---
```

```
FSE1=-8*p.q;
```

```
Out[101]: 
$$-8(p \cdot q)$$

```

Notice that you never had to write Vectors `p,q`; or Indices `mu`;. feynsage found them in your expression. On a laptop with an Apple M2 chip a trace with 10395 terms takes 0.08 s in total. FORM alone needs 0.014 s of that. The rest is the time Sage needs to build such a long formula, so for big problems it pays to put in the kinematics with `rules=` and get a short answer back.

18 When something goes wrong

Errors are written in the language of physics. Three common mistakes:

```
In [102]: for bad in (lambda: slash(p + m),
                    lambda: dirac_trace(gamma(var('muu')) * gamma(nu)),
                    lambda: contract(gamma(mu) * gamma(mu) * gamma(mu))):
    try:
        bad()
    except Exception as err:
        print(type(err).__name__ + ":", str(err).splitlines()[0])
```

```
TypeError: m + p: the part m is not a momentum. Write slash(p) + m, not slash(p + m)
CompileError: gamma(muu) is Euler's Gamma function of the symbol muu here, because muu is
not a Lorentz index. Declare it first: muu = lorentz_indices('muu')
DiracError: the index mu appears more than twice in one term; an index is either free
(once) or summed (twice). Rename one pair.
```

The first is `slash(p + m)`, which should be `slash(p) + m`. The second is a misspelled index. The third has the index μ three times in one term, which has no meaning in the summation convention.

19 How the results in this guide were checked

Every formula from the book quoted here was read on the printed page of Peskin and Schroeder [1], in the 2018 CRC Press printing: eq. (3.68) on p. 50, eqs. (5.5) and (5.6) on p. 134, eqs. (5.9) and (5.10) on p. 135, eqs. (5.11) to (5.13) on p. 136, eq. (5.87) on p. 162, eqs. (5.89) to (5.92) on p. 163 problems 5.5 and 5.6 on pp. 173 and 174 and problems 6.2 and 6.3 on pp. 209 and 210. The equations of Applications 5 to 7 were compared with Xianyu's solution [2], eqs. (5.80) to (5.93) on pp. 45 to 47 and eqs. (6.12) to (6.40) on pp. 53 to 57, read on the page. All final results agree with feynsage. The steps of problem 6.2 that do not follow from each other are listed in Application 6. The ratio R and the two-gluon colour factor were derived here and not taken from a book.

The test file `tests/test_qft_dirac.sage` runs 93 checks. It compares traces with explicit 4×4 gamma matrices in Minkowski and Euclidean space (all 256 index values of the γ^5 trace), with FORM programs run directly, with the equations above and with explicit Gell-Mann matrices for colour. It also checks that the wrong inputs of Section 18 are refused.

References

- [1] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory* (Westview Press, 1995).

- [2] Z.-Z. Xianyu, A Complete Solution to Problems in "an Introduction to Quantum Field Theory" by Peskin and Schroeder, 2016.